

VLADEASA 1800, Romania

WMO: 151190

Lat: 46.7592N

Lon: 22.7942E

Elev: 1837

StdP: 81.12

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.7	(c) -15.4	(d) -27.1	(e) 0.4	(f) -6.3	(g) -22.1	(h) 0.6	(i) -11.4	(j) 33.9	(k) -8.7	(l) 28.2	(m) -8.9	(n) 6.1	(o) 270	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.7	(c) 19.6	(d) 14.0	(e) 18.1	(f) 13.3	(g) 16.6	(h) 12.6	(i) 15.1	(j) 18.1	(k) 14.2	(l) 16.9	(m) 13.4	(n) 15.9	(o) 4.6	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	12.5	16.1	13.1	11.8	15.4	12.2	11.1	14.6	49.1	18.2	46.2	16.9	43.7	16.0	17.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.8	(b) 20.3	(c) 17.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-6.8	-5.2	-4.0	0.8	6.2	10.1	11.0	12.1	7.1	3.6	-0.4	-4.9	(6)
(7)		DBStd	7.88	4.69	4.21	4.40	5.43	3.70	3.72	3.37	3.67	4.69	4.90	5.08	4.54	(7)
(8)		HDD10.0	2972	521	425	433	277	127	46	29	22	109	206	313	463	(8)
(9)		HDD18.3	5786	779	658	691	527	378	248	229	196	337	458	563	722	(9)
(10)		CDD10.0	230	0	0	0	1	8	48	58	86	23	6	1	0	(10)
(11)		CDD18.3	2	0	0	0	0	0	0	0	1	1	0	0	0	(11)
(12)		CDH23.3	1	0	0	0	0	0	0	0	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	6.9	7.9	6.9	6.2	5.1	4.8	4.7	3.9	5.1	5.2	6.8	6.8	(14)
(15)	Precipitation	PrecAvg	863	46	43	52	62	87	121	114	75	81	64	58	62	(15)
(16)		PrecMax	1184	93	101	138	98	152	211	218	159	227	183	151	135	(16)
(17)		PrecMin	587	16	12	12	13	33	24	47	4	14	3	0	9	(17)
(18)		PrecStd	153	20	22	31	22	34	45	49	36	44	41	30	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	6.5	8.1	13.5	16.6	19.7	20.7	22.2	20.7	15.6	11.6	5.8	(19)
(20)			MCWB	-0.4	1.7	3.3	7.6	10.1	14.3	14.1	15.2	13.4	8.8	2.6	0.7	(20)
(21)		2%	DB	1.2	3.4	5.7	11.2	14.0	18.2	19.1	19.9	16.9	13.5	9.2	3.5	(21)
(22)			MCWB	-1.0	1.3	2.4	6.2	9.8	14.0	14.1	14.1	12.0	7.5	3.9	0.6	(22)
(23)		5%	DB	0.3	1.1	3.7	9.2	12.5	16.7	17.3	18.1	14.4	11.6	7.3	2.1	(23)
(24)			MCWB	-0.9	-0.7	0.9	5.2	9.0	13.2	13.0	13.3	10.8	6.9	3.6	-0.1	(24)
(25)		10%	DB	-0.6	-0.1	2.0	7.6	11.0	15.3	15.6	16.5	12.8	9.9	5.9	0.8	(25)
(26)			MCWB	-1.4	-0.9	0.1	4.4	8.1	12.5	12.3	12.6	10.0	6.2	3.1	-0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.9	2.7	4.7	8.2	11.5	16.3	15.7	16.4	13.8	9.8	6.8	2.4	(27)
(28)			MCDWB	1.3	4.4	7.2	12.1	14.0	18.0	18.9	20.7	18.3	13.3	8.9	3.5	(28)
(29)		2%	WB	0.2	1.5	3.0	7.0	10.3	14.9	14.7	15.0	12.4	9.0	5.9	1.4	(29)
(30)			MCDWB	0.8	3.0	5.2	10.4	13.2	16.9	17.5	18.6	15.7	11.7	7.4	2.6	(30)
(31)		5%	WB	-0.4	0.2	1.5	6.0	9.3	13.8	13.8	14.0	11.4	8.1	4.9	0.5	(31)
(32)			MCDWB	0.2	0.8	3.4	8.7	11.8	16.1	16.3	16.9	14.2	10.5	6.2	1.5	(32)
(33)		10%	WB	-1.3	-0.6	0.2	4.8	8.3	12.8	12.9	13.2	10.4	7.0	3.8	-0.3	(33)
(34)			MCDWB	-0.7	-0.2	1.6	7.2	10.3	14.8	15.3	15.9	12.6	9.0	5.1	0.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.8	5.0	5.3	5.3	5.4	5.6	5.7	5.2	5.0	4.4	4.5	(35)	
(36)			MCDBR	4.3	5.3	7.0	7.4	6.7	7.1	7.7	7.1	7.0	6.5	5.6	5.2	(36)
(37)			MCWBR	4.0	3.9	5.2	4.9	5.1	5.3	5.5	4.7	4.6	5.2	4.0	3.9	(37)
(38)		5% WB	MCDWR	4.2	4.7	6.6	7.2	6.4	6.7	6.6	6.3	6.2	5.9	4.7	4.9	(38)
(39)			MCWBR	4.0	4.1	5.2	4.8	4.9	5.3	5.2	4.8	4.3	5.1	4.0	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.263	0.286	0.320	0.360	0.349	0.346	0.358	0.361	0.333	0.308	0.286	0.264	(40)	
(41)		taud	2.414	2.383	2.350	2.267	2.340	2.366	2.326	2.317	2.386	2.457	2.473	2.452	(41)	
(42)		Ebn at Noon	847	892	905	894	915	917	899	882	879	854	809	803	(42)	
(43)		Edh at Noon	68	86	105	125	121	118	122	118	100	80	64	59	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.16	1.97	3.14	4.37	5.29	5.83	5.71	5.26	3.72	2.51	1.43	0.90	(44)	
(45)		RadStd	0.17	0.27	0.46	0.46	0.51	0.46	0.50	0.49	0.43	0.30	0.20	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)		+0.71	N/A	N/A	+0.98	N/A	N/A	-127	-191	+113	+57

Nomenclature: See separate page